Omer Muhammad Javed

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ARTIFICIAL INTELLIGENCE & ROBOTICS | AUTONOMOUS SYSTEMS | COMPUTER VISION

Profile

First Class Artificial Intelligence graduate currently studying an MSc in Robotics at the University of Manchester. Hands-on experience developing robotics, computer vision and autonomous systems, with practical experience integrating machine learning, real-time perception, navigation and drone control. Developed an end-to-end autonomous drone system for construction-site safety using YOLOv8, OpenCV, PX4 and MAVSDK, alongside experience with robotics simulation and software development. Seeking graduate and entry-level opportunities in Robotics, Robotics Software, Software Engineering, Computer Vision and Autonomous Systems.

Technical Skills

Programming: Python, Java, C#, SQL
Robotics: ROS 2, PX4, MAVSDK, MAVLink, Gazebo, QGroundControl
AI & Computer Vision: PyTorch, YOLOv8, OpenCV, Machine Learning, Object Detection
Development: Git, Linux/Ubuntu, Tkinter
Concepts: SLAM, Autonomous Systems, Real-Time Systems, System Integration

Projects

AI-Powered Autonomous Drone for Construction Site PPE Detection | Individual Capstone Project (University of Essex)

- Developed an end-to-end autonomous drone system for construction-site safety monitoring.
- Trained YOLOv8 object detection model achieving 0.93 mAP@0.5 for hardhat compliance detection.
- Implemented autonomous mission and teach-and-repeat navigation using PX4 and MAVSDK.
- Developed modular Python architecture integrating perception, mission control and telemetry.
- Built real-time OpenCV/YOLOv8 detection pipelines and a Tkinter monitoring dashboard.
- Tested system using SITL simulation and controlled real-world environments.

Autonomous Mobile Robot Navigation | Mini Assignment Project (University of Essex)

- Developed a simulated mobile robot capable of autonomous navigation between target locations using SLAM.
- Generated an environment map and used localisation/navigation techniques to support autonomous movement.

Child Health Tracking Application | Group Project (University of Essex)

- Developed a Java Swing application for recording sleep, feeding, medication and other daily health data.
- Implemented CSV data persistence, daily report generation and GUI functionality.

Education

MSc Robotics | University of Manchester | September 2026 – Present

BSc Artificial Intelligence | University of Essex | October 2023 – June 2026
First Class Honours
Relevant technical areas: Machine Learning, Mobile Robotics, Natural Language Engineering, Data Science, Artificial Intelligence, Software Development

Experience

Rental Property Manager | Airbnb, Bradford | August 2022 – July 2025

- Managed day-to-day operations including bookings, scheduling, guest communication and issue resolution.
- Worked independently across multiple concurrent responsibilities while maintaining service standards.

Data Entry / Operations Support | HiTech Powder Coating, Leeds | January 2020 – March 2022

- Maintained operational records and supported administrative processes using Excel.
- Developed strong attention to detail, organisation and data-management skills.

Additional Information

Full UK driving licence | Open to relocation